PHYSICS

1)

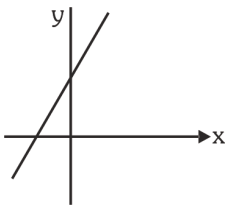
Find the value of
$5(\sin100^{o}\cos27^{o} + \cos100^{o}\sin27^{o})$

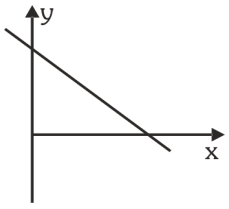
(1) 1
(2) 2
(3) 3
(4) 4

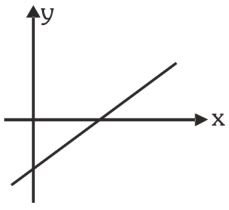
2) Value of cos106° is:

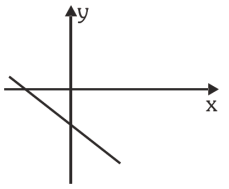
(1) $\frac{7}{25}$
(2) $\frac{-7}{25}$
(3) $\frac{24}{25}$
(4) $\frac{-24}{25}$

3) Graph of equation 3x – 2y + 4 = 0 is–

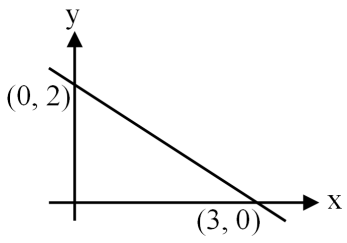
(1) 

(2) 

(3) 

(4) 

4) Calculate slope of shown line



(1) $\frac{3}{2}$

(2) $\frac{2}{3}$

(3) $-\frac{3}{2}$

(4) $-\frac{2}{3}$

5) $\frac{d}{dx}(x^5 + x^4 + 7)$ equals to :

(1) $x^6 + x^5 + 7x$

(2) $5x^4 + 4x^3 + 7$

(3) $\frac{x^6}{6} + \frac{x^5}{5} + 7x$

(4) $5x^4 + 4x^3$

6) If $y = 6x^3 - 4x$ then the value of $\frac{d^2y}{dx^2}$ at x = 1.

(1) 36

(2) 18

(3) 0

(4) 72

7) Find the minimum value of y, if $y = x^2 + 4x$

(1) –4

(2) 4

(3) –2

(4) 0

8) A metallic disc is being heated. Its area (in m^2) at any time t (in sec) is given by $A = 5t^2 + 4t$. Calculate the rate of increase in area at t = 3sec.

(1) 4 m^2/s

(2) 34 m^2/s

(3) 57 m^2/s

(4) 63 m^2/s

9) Integrate the function with respect to x : $\int \left(4\sin x - \frac{2}{x}\right) dx$

(1) –4cosx – 2nx + c

(2) 4cosx + 2nx + c

(3) –4cosx + 2nx + c

(4) None of these

10) Determine the average value of y = 2x +3 in the interval $0 \le x \le 1$:-

(1) 1

(2) 5

(3) 3

(4) 4

11) The roots of equation $2x^2 - x - 3 = 0$ are:

(1) $\frac{3}{2}, -1$

(2) $-\frac{3}{2}, 1$

(3) $\frac{2}{3}, \frac{5}{2}$

(4) $\frac{3}{2}, -\frac{5}{2}$

12) Sum of the series :-
S = 4 + 8 + 12 + ... + 64

(1) 464

(2) 540

(3) 544

(4) 646

13)

Take $\log 2 = 0.3$ and $\log 3 \equiv 0.5$. Find log 18

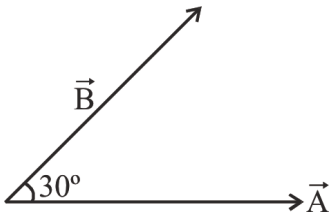
(1) 0.8

(2) 1.3

(3) 1

(4) 1.1

14) If angle between $\vec{A}$ and $\vec{B}$ is 30°, then find out the angle between $3\vec{A}$ and $-\frac{2}{3}\vec{B}$.



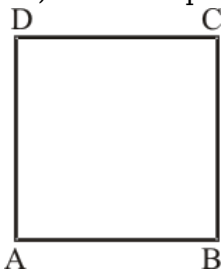
(1) 180°

(2) 150°

(3) 90°

(4) 30°

15) For the square shown :-



(1) $\vec{AB} + \vec{AD} = \vec{AC}$

(2) $\vec{BA} + \vec{AD} = \vec{BD}$

(3) $\vec{AB} + \vec{BC} + \vec{CD} - \vec{AD} = \vec{0}$

(4) All of the above

16) Two equal forces are acting at a point with an angle of 60° between them. If the resultant force is equal to $40\sqrt{3}N$, the magnitude of each force is :-

(1) 40N

(2) 20N

(3) 80N

(4) 30N

17) Given that $\vec{A} + \vec{B} = \vec{C}$ and that $\vec{C}$ is $\perp$ to $\vec{A}$. Further if $|\vec{A}| = |\vec{C}|$, then what is the angle between $\vec{A}$ and $\vec{B}$:-

(1) $\frac{\pi}{4}$ radian

(2) $\frac{\pi}{2}$ radian

(3) $\frac{3\pi}{4}$ radian

(4) π radian

18)

If $|\vec{A} - \vec{B}| = |\vec{A}| = |\vec{B}|$, the angle between $\vec{A}$ and $\vec{B}$ is :

(1) 60º

(2) 0º

(3) 120º

(4) 90º

19) If the angle between the unit vectors $\hat{a}$ and $\hat{b}$ is 60°, then $\left|\hat{a}-\hat{b}\right|$ is :-

(1) 0

(2) 1

(3) 2

(4) $\sqrt{3}$

20) What should be added to the sum of vectors $(\hat{i}-2\hat{j}+2\hat{k})$ and $(2\hat{i}+\hat{j}-\hat{k})$ so that the resultant is a unit vector along the x-axis ?

(1) $2\hat{i}+\hat{j}-\hat{k}$

(2) $-2\hat{i}+\hat{j}-\hat{k}$

(3) $2\hat{i}-\hat{j}+\hat{k}$

(4) $-2\hat{i}-\hat{j}-\hat{k}$

21) For two vectors $\vec{A}$ and $\vec{B}$, sum $\left(\vec{A}+\vec{B}\right)$ is perpendicular to the difference $\left(\vec{A}-\vec{B}\right)$. The ratio of their magnitudes is :-

(1) 1

(2) 2

(3) 3

(4) None of these

22) If $\vec{A}=2\hat{i}+4\hat{j}-5\hat{k}$ the direction of cosines of the vector $\vec{A}$ are :-

(1) $\frac{2}{\sqrt{45}}, \frac{4}{\sqrt{45}}$ and $\frac{-5}{\sqrt{45}}$

(2) $\frac{1}{\sqrt{45}}, \frac{2}{\sqrt{45}}$ and $\frac{3}{\sqrt{45}}$

(3) $\frac{4}{\sqrt{45}}$, 0 and $\frac{4}{\sqrt{45}}$

(4) $\frac{3}{\sqrt{45}}, \frac{2}{\sqrt{45}}$ and $\frac{5}{\sqrt{45}}$

23) The angle between two vectors given by $6\hat{i}+6\hat{j}-3\hat{k}$ and $7\hat{i}+4\hat{j}+4\hat{k}$ is :-

(1) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$

(2) $\cos^{-1}\left(\frac{5}{\sqrt{3}}\right)$

(3) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$

(4) $\sin^{-1}\left(\frac{\sqrt{5}}{3}\right)$

24) Find the component of $\vec{A} = \hat{i} + 2\hat{j} + 2\hat{k}$ on $\vec{B} = \hat{j} + \hat{k}$:-

(1) $\frac{\hat{j} + \hat{k}}{\sqrt{2}}$

(2) $2(\hat{j} + \hat{k})$

(3) $\frac{4}{3}(\hat{j} + \hat{k})$

(4) $\frac{4}{\sqrt{2}}(\hat{j} + \hat{k})$

25) What is the unit vector perpendicular to the following vectors $2\hat{i} + 2\hat{j} - \hat{k}$ and $6\hat{i} - 3\hat{j} + 2\hat{k}$

(1) $\frac{\hat{i} + 10\hat{j} - 18\hat{k}}{5\sqrt{17}}$

(2) $\frac{\hat{i} - 10\hat{j} + 18\hat{k}}{5\sqrt{17}}$

(3) $\frac{\hat{i} - 10\hat{j} - 18\hat{k}}{5\sqrt{17}}$

(4) $\frac{\hat{i} + 10\hat{j} + 18\hat{k}}{5\sqrt{17}}$

26) A person moves 30 m north and then 20 m towards east and finally 30 $\sqrt{2}$ m in south-west direction. The displacement of the person from the origin will be :-

(1) 10 m along north

(2) 10 m long south

(3) 10 m along west

(4) Zero

27) A cyclist moving on a circular track of radius 40 m completes half a revolution in 40 second. Its average velocity is :

(1) Zero

(2) 4π m/s

(3) 2 m/s

(4) 8 π m/s

28) A body covers half the distance of its journey with speed 10 ms^{-1} and remaining half with

speed 15 ms^{-1} along a straight line. The average speed is:-

(1) 12 ms^{-1}

(2) 10 ms^{-1}

(3) 5 ms^{-1}

(4) 12.5 ms^{-1}

29) A scooter is going towards east at 10 ms^{-1} turns right through an angle of 90°. If the speed of the scooter remains unchanged in taking this turn, the change in the velocity of the scooter is :-

(1) 20.0 ms^{-1} is south-western direction

(2) zero

(3) 10.0 ms^{-1} in southern direction

(4) 14.14 ms^{-1} in south-western direction

30) A body is moving with velocity 10 m/s towards North and after 2 seconds its velocity becomes 20 m/s towards West. The average acceleration of the body will be :-

(1) $5\sqrt{5}$ m/s^2

(2) 25 m/s^2

(3) 125 m/s^2

(4) 15 m/s^2

31) If the distance covered by a particle is given by the relation $x = at^2$. The particle is moving with:

(1) constant acceleration

(2) zero acceleration

(3) variable acceleration

(4) none of these

32) A particle moves along x-axis and its displacement at any time is given by $x(t) = 2t^3 - 3t^2 + 4t$ in SI units. The velocity of the particle when its acceleration is zero, is :

(1) 2.5 m s^{-1}

(2) 3.5 m s^{-1}

(3) 4.5 m s^{-1}

(4) 8.5 m s^{-1}

33) For motion of an object along the x-axis the velocity v depends on the displacement x as $v = 3x^2 - 2x$, then what is the acceleration at x = 2m.

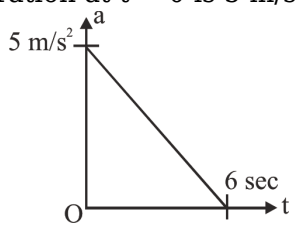
(1) 48 ms^{-2}

(2) 80 ms^{-2}

(3) 18 ms^{-2}

(4) 10 ms^{-2}

34) A particle having initial speed 10 m/s. Its acceleration at t = 0 is 5 m/s^2 which varies with time as shown. The maximum speed of the particle will be :-



(1) 15 m/s

(2) 25 m/s

(3) 30 m/s

(4) 7.5 m/s

35)

Two cars P and Q start from a point at the same time in a straight line and their positions are represented by $x_p(t) = at + bt^2$ and $x_Q(t) = ft - t^2$. At what time do the cars have the same velocity ?

(1) $\frac{a+f}{2(1+b)}$

(2) $\frac{f-a}{2(1+b)}$

(3) $\frac{a-f}{1+b}$

(4) $\frac{a+f}{2(b-1)}$

36) The initial velocity of a particle is given by u (at t = 0) and the acceleration by f, where f = at (here t is time and a is constant). Which of the following relation is valid?

(1) $v = u + at^2$

(2) $v = u + \frac{at^2}{2}$

(3) v = u + at

(4) v = u

37) A particle has an velocity $4\hat{i} + 3\hat{j}$ and an acceleration of $0.3\hat{i} + 0.4\hat{j}$, its speed after 10 sec is :-

(1) 10

(2) $7\sqrt{2}$

(3) 8.5

(4) 7

38) The initial velocity of the particle is 10m/sec and its retardation is 2m/sec^2. The distance moved by the particle in 5th second of its motion is :-

(1) 1m

(2) 19m

(3) 50m

(4) 75m

39) A stone falls freely such that the distance covered by it in the last second of its motion is equal to the distance covered by it in the first 5 sec. It remained in air for :-

(1) 12 sec

(2) 26 sec

(3) 13 sec

(4) 25 sec

40) Water drops are falling from roof at regular time interval such that first drop reaches the ground, 4th drop is about to leaves the tap. If height of the tap is 99 m then find out height 3rd drop from the ground ?

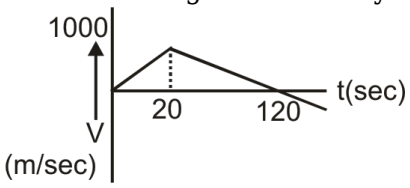
(1) 11 m

(2) 55 m

(3) 88 m

(4) 90 m

41) A rocket is projected vertically upwards and its time velocity graph is shown in the figure. The maximum height attained by the rocket is



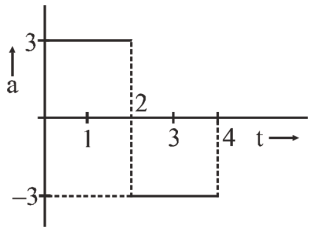
(1) 1 km

(2) 10 km

(3) 100 km

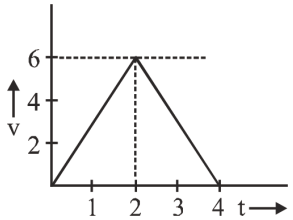
(4) 60 km

42) A particle starts from rest at t = 0 and undergoes an acceleration a in ms^{-2} with time t in seconds which is as shown:

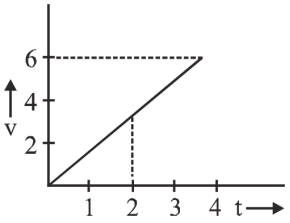


Which one of the following plot represents velocity v in ms^{-1} versus time t in seconds?

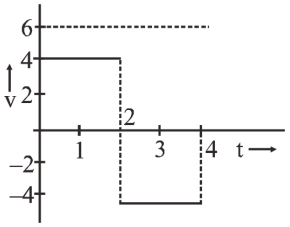
(1)



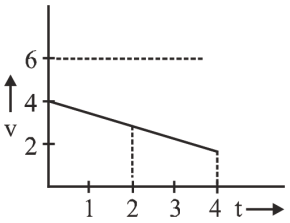
(2)



(3)



(4)



43) A body is projected at 60° with ground. It covers a horizontal distance of 100 m. If the same body is projected at 60° with vertical with same velocity, the new range is :-

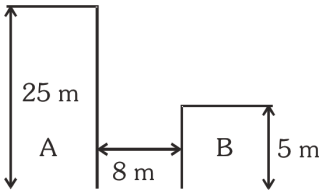
(1) 50 m

(2) 100 m

(3) 200 m

(4) 150 m

44) A boy wants to jump from building A to building B. Height of building A is 25 m and that of building B is 5m. Distance between buildings is 8m. Assume that the boy jumps horizontally, then calculate minimum velocity with which he has to jump to land safely on building B.



(1) 6 m/s

(2) 8 m/s

(3) 4 m/s

(4) 2 m/s

45) A 120 m long train is moving in a direction with speed 20 m/s. Another train B moving with 30 m/s in the opposite direction and 130 m long crosses the first train in a time :

(1) 6 s

(2) 25 s

(3) 10 s

(4) 5 s

CHEMISTRY

1) **Assertion (A) :-** From the value of $|\Psi|^2$ at different points within an atom, it is possible to predict the region around the nucleus where electron will most probably be found.
Reason (R) :- With the help of quantum mechanical model of atom, exact position of an electron in an atom can be determined.

(1) Both assertion and reason are correct and reason is the correct explanation of assertion

(2) Both assertion and reason are correct but reason is not the correct explanation of assertion

(3) Assertion is correct and reason is incorrect

(4) Both assertion and reason are incorrect

2) Given below are two statements:
Statement-I : The energy of He^+ ion in n = 2 state is same as the energy of H atom in n = 1 state.
Statement-II : It is possible to determine simultaneously the exact position and exact momentum of an electron in H atom.
In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **Statement I** and **Statement II** are false.

(2) **Statement I** is true but **Statement II** is false.

(3) **Statement I** is false but **Statement II** is true.

(4) Both **Statement I** and **Statement II** are true.

3) For copper atom match the column :

Column-I		**Column-II**	
(a)	Number of e^- with n = 4, ℓ = 0, m = 0	(i)	28
(b)	Number of e^- with n = 3, $s = +\frac{1}{2}$	(ii)	18
(c)	Number of e^- in penultimate shell	(iii)	9
(d)	Number of e^- in its kernel	(iv)	1

(1) (a)→(i), (b)→(ii), (c)→(iii), (d)→(iv)

(2) (a)→(iv), (b)→(iii), (c)→(i), (d)→(ii)

(3) (a)→(iv), (b)→(iii), (c)→(ii), (d)→(i)

(4) (a)→(iv), (b)→(ii), (c)→(iii), (d)→(i)

4)

How many subshells and electron are associated with n = 4

(1) 4, 32

(2) 4, 16

(3) 4, 8

(4) 4, 4

5)

Correct order of specific charge is -

(1) e > α > n > P

(2) n > P > α > e

(3) P > α > n > e

(4) e > P > α > n

6) If mass of neutron is doubled, mass of proton is halved and mass of electron is doubled, find out the % change of atomic weight of C^{12}

(1) 50%

(2) 25%

(3) 60%

(4) 75%

7) The kinetic and potential energy of e^- in second Bohr's orbit of hydrogen atom are respectively:-

(1) +6.8 eV and –3.4 eV

(2) –1.5 eV and +3.02 eV

(3) –6.8 eV and +3.4 eV

(4) +3.4 eV and – 6.8 eV

8) Which of the following expressions about Bohr's atomic model is/are correct?

(1) $E_n = -2.18 \times 10^{-18} \times \frac{Z^2}{n^2}$ J/atom

(2) $\Delta E = 2.18 \times 10^{-18} \times Z^2 \left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right)$ J

(3) $r_n = 52.9 \times \frac{n^2}{Z}$ pm

(4) All of the above

9) Which of the following orbital has xy and yz nodal plane

(1) d_{xy}

(2) d_{yz}

(3) d_{zx}

(4) dz^2

10) If the shortest wavelength of Lyman series of H atom is x, then the wave length of Ist line of Balmer series of H atom will be :-

(1) $\frac{9x}{5}$

(2) $\frac{36x}{5}$

(3) $\frac{5x}{9}$

(4) $\frac{5x}{36}$

11)

(i)	${}_{26}Fe^{54}$, ${}_{26}Fe^{56}$, ${}_{26}Fe^{57}$, ${}_{26}Fe^{58}$	(a)	Isotopes
(ii)	${}_{1}H^{3}$, ${}_{2}He^{3}$	(b)	Isotones
(iii)	${}_{32}Ge^{76}$, ${}_{33}As^{77}$	(c)	Isodiaphers
(iv)	${}_{92}U^{235}$, ${}_{90}Th^{231}$	(d)	Isobars
(v)	${}_{1}H^{1}$, ${}_{1}D^{2}$, ${}_{1}T^{3}$		

Match the above correct terms:-

(1) [(i), - a], [(ii) - d], [(iii) - b], [(iv) - c], [(v) - a]

(2) [(i) - a] [(ii) - d], [(iii) - d] [(iv) - c] [v - a]

(3) [v -a] [(iv) - c]. [(iii) - d] [(ii) - b] [(i) - a]

(4) None of them

12) Two particles A and B are in motion, wavelength related to particle A is 5×10^{-8} m. If the momentum of particle B is half of A then what will be the wavelength of B :-

(1) 2.5×10^{-8} m

(2) 1.25×10^{-8} m

(3) 1×10^{-7} m

(4) 1×10^{-8} m

13) Ratio of frequency of revolution of electron in the second excited state of He^{+} and second state of hydrogen atom is :-

(1) $\frac{32}{27}$

(2) $\frac{27}{32}$

(3) $\frac{1}{54}$

(4) $\frac{27}{2}$

14) A 100 W bulb emits light of wavelength 400 nm number of photons emitted per second by the bulb is :

(1) 2.0×10^{20}

(2) 4.0×10^{19}

(3) 6.0×10^{20}

(4) 8.0×10^{-20}

15) Which of the following statement is true :-
(a) 6 electrons are present in Mg for which m = 0
(b) Maximum 6 electrons can accomdate in p_x-orbital
(c) Maximum 18 electrons present in M-shell
(d) 3–electron present in phosphorous for which,
l = 0, S = $+\frac{1}{2}$

(1) a, b, c, d

(2) a, b c

(3) a, c, d

(4) c, d

16) **Statement-I** : For any gas, vapour density = $\frac{\text{molecular weight}}{2}$ so unit of vapour density is g/mole.
Statement-II : Molar volume of an ideal gas at STP is 22.4 L.

(1) Statement-I and statement-II both are correct.

(2) Statement-I and statement-II both are incorrect.

(3) Statement-I is correct and statement-II is incorrect.

(4) Statement-I is incorrect and statement-II is correct.

17) **Assertion :** The number of O atoms in 16 g of oxygen & 16 g of ozone is same.
Reason : Each of the above species represent 1 g - molecule of oxygen.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

18) Match the following Column-I to Column-II (Volume at STP).

Column-I		**Column-II**	
(A)	14 g of nitrogen gas	(P)	22.4 L
(B)	6.02×10^{22} molecules of NH_3	(Q)	4.48 L
(C)	1 mole of SO_2	(R)	11.2 L

(D)	0.2 mole of O_3	(S)	2.24 L

(1) (A)–(R); (B)–(P); (C)–(S); (D)–(Q)

(2) (A)–(R); (B)–(Q); (C)–(S); (D)–(P)

(3) (A)–(R); (B)–(S); (C)–(P); (D)–(Q)

(4) (A)–(Q); (B)–(P); (C)–(R); (D)–(S)

19) The number of atoms in 0.1 mol of a triatomic gas is :

(1) 1.800×10^{22}

(2) 6.026×10^{22}

(3) 1.806×10^{23}

(4) 3.600×10^{22}

20) An organic compound contains 4% sulphur. Its minimum molecular weight is :

(1) 200

(2) 400

(3) 800

(4) 1600

21) How many mL of 0.1 M aqueous solution of NaOH contain 4g of NaOH :-

(1) 1 mL

(2) 100 mL

(3) 1000 mL

(4) 500 mL

22) What amount of CaO will be produced by 2 moles of Ca.
$2Ca + O_2 \rightarrow 2CaO$

(1) 5.6g

(2) 56g

(3) 112g

(4) 224g

23) 100 mL of gaseous hydrocarbon on combustion give 400 mL of $CO_{2(g)}$ and 500 ml of H_2O (vap). The hydrocarbon is :-

(1) C_4H_6

(2) C_8H_{10}

(3) C_4H_8

(4) C_4H_{10}

24) A mixture containing 64.0 g H_2 and 64.0 g O_2 is reacted so that water is formed as follows
$2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(g)}$

Which of the following statements is/are correct ?

(1) H_2 is the limiting reagent

(2) O_2 is the limiting reagent

(3) The reaction mixture contains 72.0 g of H_2O and 56.0 g of unreacted H_2 after the reaction.

(4) Both (2) and (3)

25) One mole of P_4 molecule contain :-

(1) 1 molecule

(2) 4 molecules

(3) $\frac{1}{4} \times N_A$ atoms

(4) 24.08×10^{23} atoms

26) 3 L of mixture of propane (C_3H_8) & butane (C_4H_{10}) on complete combustion gives10 L CO_2. Find the composition of mixture.

(1) C_3H_8, 2L and C_4H_{10}, 1L

(2) C_3H_8, 3L and C_4H_{10}, 0L

(3) C_3H_8, 1.5L and C_4H_{10}, 1.5L

(4) C_3H_8, 0L and C_4H_{10}, 3L

27) Molecular weight of SO_2 will be

(1) 64g

(2) 64 amu

(3) 64

(4) All

28) What is the percentage purity of a sample of $CaCO_3$ if 150 g of the sample gives out 44g of CO_2 gas on heating :-

(1) 33.33%

(2) 50%

(3) 66.67%

(4) 75%

29) A compound contains 40% carbon, 6.7% hydrogen and 53.3% oxygen by mass. Find its empirical formula :-

(1) C_2H_4O

(2) CH_2O

(3) CH_4O

(4) C_2H_2O

30) Relation between molarity(M) and molality(m) is : (d = density and M_{solute} = Molar mass)

(1) $m = \frac{1000 \times M}{1000 \times d - M}$

(2) $M = \frac{1000 \times M}{1000 \times d - mM}$

(3) $m = \frac{M}{1000 \times d - M \times M_{Solute}}$

(4) $m = \frac{1000 \times M}{(1000 \times d) - (M \times M_{Solute})}$

31) **Statement-I :** For the physical equilibrium $H_2O(s) \rightleftharpoons H_2O(\square)$ on increasing temperature and increasing pressure more water will form.
Statement-II : Since forward reaction is endothermic in nature and volume of water is greater than that of volume of ice.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

32)

Match the following columns :

	Column-I (Chemical reaction)		**Column-II (Relation between K_P and K_C)**
(i)	$C_{(s)} + CO_{2(g)} \rightleftharpoons 2CO_{(g)}$	(a)	$K_P = K_C$
(ii)	$N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$	(b)	$K_P = K_C(RT)$
(iii)	$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$	(c)	$K_P = K_C(RT)^{-1}$
(iv)	$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$	(d)	$K_P = K_C(RT)^{-2}$

(1) (i)–(b), (ii)–(a), (iii)–(d), (iv)–(c)

(2) (i)–(d), (ii)–(a), (iii)–(b), (iv)–(c)

(3) (i)–(a), (ii)–(b), (iii)–(d), (iv)–(c)

(4) (i)–(c), (ii)–(a), (iii)–(d), (iv)–(b)

33) **Assertion (A)** :- For $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$. If more Cl_2 is added the equilibrium will shift in backward direction hence equilibrium constant will decrease. **Reason (R)** :- Addition of inert gas to the equilibrium mixture at constant volume does not alter the equilibrium.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

34) Equilibrium constant unit of for this gaseous reaction $4A + 5B \rightleftharpoons 4X + 6Y$

(1) $mol^2\ L^2$

(2) $L\ mol^{-1}$

(3) $L^2\ mol^{-2}$

(4) $mol\ L^{-1}$

35) For a chemical reaction of the type $A \rightleftharpoons B$, K = 2.0 and $B \rightleftharpoons C$, K = 0.01. Equilibrium constant for the reaction $2C \rightleftharpoons 2A$ is :-

(1) 25

(2) 50

(3) 2.5×10^3

(4) 4×10^{-4}

36) Given a gas phase reaction,
$2A(g) + B(g) \rightleftharpoons C(g) + D(g)$
Which one of the following changes will affect the value of K_C ?

(1) Addition of inert gas

(2) Addition of catalyst

(3) Addition of reactants

(4) Increasing temperature

37) For which of the following reaction value of $\frac{K_p}{K_c}$ is one :-

(1) $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$

(2) $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

(3) $SO_{2(g)} + \frac{1}{2}O_{2(g)} \rightleftharpoons SO_{3(g)}$

(4) $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$

38) Relation $\log \frac{K_c}{K_p} + \log RT = 0$ is true for :-

(1) $2A(g) \rightleftharpoons B(g) + 2C(g)$

(2) $2A(g) + B(s) \rightleftharpoons 2C(g)$

(3) $2A(s) + B(g) \rightleftharpoons C(g)$

(4) $A(g) + B(g) \rightleftharpoons C(g)$

39) Densities of diamond and graphite are 3.5 and 2.3 g/mL. $C(diamond) \rightleftharpoons C(graphite)$; $\Delta_r H = -1.9$ kJ/mol favourable conditions for formation of graphite are :-

(1) high pressure and low temperature.
(2) low pressure and high temperature.
(3) high pressure and high temperature.
(4) low pressure and low temperature.

40)

Find the active mass of 50 gms of $CaCO_3$ in 1 L vessel :-

(1) 0.5 mol/L
(2) 0.2 mol/L
(3) 1
(4) 0.25 mol/L

41) For an equilibrium reaction if the value of $K_c >> 1$, then the reaction favoured more towards:

(1) Backward direction
(2) Forward direction
(3) Equilibrium will be obtained
(4) Reaction will stop

42) Match the following:

Column-I		**Column-II**	
a	Q=K	p	Reaction is nearer to completion
b	Q<K	q	Reaction is not at equilibrium
c	Q>K	r	Reaction proceeds in forward direction
d	K>>1	s	Reaction is in equilibrium
		t	Reaction proceeds in backward direction

(1) a-s ; b-q,r ; c-q,t ; d-p
(2) a-s ; b-p,t ; c-t ; d-p
(3) a-q ; b-r ; c-p,t ; d-q
(4) a-r ; b-q ; c-t ; d-q,t

43)

At 527°C, the reaction given below has $K_C = 4$

$$NH_3(g) \rightleftharpoons \frac{1}{2} N_2(g) + \frac{3}{2} H_2(g)$$

What is the K_p for the reaction ?

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

(1) $16 \times (800\ R)^2$

(2) $\left(\frac{800R}{4}\right)^{-2}$

(3) $\left(\frac{1}{4 \times 800R}\right)^{2}$

(4) None of these

44) For the equilibrium $SO_2Cl_{2(g)} \rightleftharpoons SO_{2(g)} + Cl_{2(g)}$. What is temperature at which $\frac{K_p[atm]}{K_c[M]} = 3$.

(1) 0.027K

(2) 0.36K

(3) 36.54K

(4) 273K

45) $NaHCO_3$ decomposes as :
$2NaHCO_3(s) \rightleftharpoons Na_2CO_3(s) + CO_2(g) + H_2O(g)$
The equilibrium pressure is 1.04 atm. The K_P for the reaction is :

(1) 0.2704

(2) 2.704

(3) 27.04

(4) 270.4

BIOLOGY

1) Mast cells occur in

(1) Connective tissue

(2) Epithelial tissue

(3) Muscular tissue

(4) Nervous tissue

2) Example of loose connective tissue is

(1) Areolar

(2) Bone

(3) Blood

(4) Cartilage

3) Cartilage present in trachea, larynx and bronchi is

(1) Fibrous

(2) Elastic

(3) Hyaline

(4) Calcified

4) Collagen and elastin are formed by

(1) Macrophages
(2) Fibroblasts
(3) Mast cells
(4) Chondrocytes

5) Which is not a component of areolar tissue

(1) Macrophage
(2) Plasma cell
(3) Schwann cell
(4) Adipose cell.

6) Nasal septum gets damaged. Its recovery requires cartilage called :

(1) Fibrous cartilage
(2) Elastic cartilage
(3) Hyaline cartilage
(4) Calcified cartilage

7) What will happen if ligaments are cut or broken :-

(1) Bones will move freely at joints
(2) No movement at joint
(3) Bone will become unfix
(4) Bone will become fixed

8) **Assertion :** Epithelium tissue provides a covering or a lining for some part of the body.
Reason : Epithelium tissue has after surface which faces either body fluid or the outside environment.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.
(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(3) **Assertion** is true but **Reason** is false.
(4) **Assertion** is false but **Reason** is true.

9) **Assertion (A) :** Bone is the main tissue that provides structural framework of the body. Bones support and protect soft tissues and organs.
Reason (R) : Bone have a hard and non-pliable ground substance rich in calcium salts and collagen fibers which given bone it's strength.

(1) Both A and R are true and R is correct explanation of A
(2) Both A and R are true but R is not correct explanation of A
(3) A is true but R is false
(4) Both A and R are false

10) Which of the following types of epithelium tissue are found in the lining of blood vessels and alveoli ?

(1) Simple cuboidal epithelium

(2) Simple squamous epithelium

(3) Startified squamous epithelium

(4) Ciliated columnar epithelium

11) **Statement-I :** Compound epithelium is made of more than one layer of cells they cover the moist surface of the skin, pharynx, inner lining of ducts of salivary glands and of pancreatic ducts.
Statement-II : Connective tissue are most distributed in the body of complex animals. They are named connective tissue because of their special function of linking and supporting other tissues/organ of the body.

(1) Both Statement-I and Statement-II are correct

(2) Both Statement-I and Statement-II are incorrect

(3) Statement-I is correct but Statement-II is incorrect

(4) Statement-I is incorrect but Statement-II is correct

12) Match the following column :

Column-I (Tissue)		**Column-II (Cell shape)**		**Column-III (Function)**		**Column-IV (Example)**	
(1)	Simple squamous epithelium	(A)	Flat cells	(W)	Diffusion	(i)	Renal tubules
(2)	Simple cuboidal epithelium	(B)	Cube shaped	(X)	Secretion and absorption	(ii)	Intestine
(3)	Simple columnar epithelium	(C)	Pillar like	(Y)	Protection	(iii)	Alveoli of lungs
(4)	Compound epithelium	(D)	Multilayered	(Z)	Absorption	(iv)	Skin

Choose correct option :-

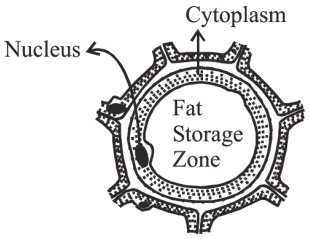
(1) 1-A, W-iii, 2-B-X-i, 3-C-Z-ii, 4-D-Y-iv

(2) 1-B, W-ii, 2-A-X-iii, 3-C-Z-iv, 4-D-Y-iv

(3) 1-A, Z-i, 2-B-W-iii, 3-C-X-ii, 4-D-Y-iv

(4) 1-D, Y-iv, 2-B-X-ii, 3-A-W-iii, 4-C-Z-i

13) Identify the following diagram and choose correct option where it makes major component :-



(1) Dense conective tissue

(2) Skeleton CT.

(3) Adipose tissue

(4) Areolar tissue

14) Endoskeleton of vertebrate embryo is made by:-

(1) Bones

(2) Cartilage

(3) Ligaments and Tendons

(4) Both (2) and (3)

15) How many of the following are the functions of connective tissue
(i) To protect the delicate organs
(ii) To prepare foundation for epithelium
(iii) Deposition of fat
(iv) Functional coordination

(1) One

(2) Two

(3) Three

(4) Four

16) Most expanded and abundent tissue in our body is :-

(1) Epithelium

(2) Connective tissue

(3) Muscles

(4) Nervous tissue

17) Connective tissue which is found at tip of nose and ear pinna is :-

(1) Bone

(2) Cartilages

(3) Ligament

(4) Tendon

18) Cheif property of matrix of Bone is :-

(1) Collagen fibres are present

(2) Contain cells
(3) Contain calcium salts
(4) Contain blood vessels

19) Inner most layer of salivary duct and pancreatic duct is made up of -

(1) Simple Columner Epithelium
(2) Simple cuboidal Epithelium
(3) Simple squamous Epithelium
(4) Stratified cuboidal Epithelium

20) Heparin, histamine and serotonin are secreted by-

(1) Mesenchymal cells
(2) Fibroblast
(3) Mast cells
(4) Plasma cell

21) How many of the following statements are wrong?
(i) Hyaline cartilage is found in nasal septum
(ii) Blood capillaries are not found in epithelial tissue
(iii) Hyaline cartilage is found at the junction of two long bones
(iv) Perichondrium is absent on hyaline cartilage.

(1) 4
(2) 3
(3) 2
(4) 1

22) Intervertebral disc are composed of :-

(1) Hyaline cartilage
(2) Elastic cartilage
(3) White fibrous cartilage
(4) Calcified cartilage

23) Glass like cartilage is :-

(1) Hyaline cartilage
(2) Fibro cartilage
(3) Calcified cartilage
(4) Elastic cartilage

24) Ligaments connect :-

(1) Bones with bones

(2) Muscles with muscles

(3) Bones with muscles

(4) None of them

25) Tendon is :-

(1) Modified white fibrous tissue which connects bone to bone.

(2) Modified yellow fibrous tissue which connects bone to bone.

(3) Modified white fibrous tissue which connects bone to muscle.

(4) Modified yellow fibrous tissue which connects bone to muscle.

26)

Match column-I with column-II :-

	Column-I		**Column-II**
(a)	Mosquitoes	(i)	Living fossil
(b)	Limulus	(ii)	Parapodia
(c)	Annelida	(iii)	Book gills
(d)	Aschelminthes	(iv)	H-shaped excretory structure
(e)	Platyhelminthes	(v)	Piercing and sucking type mouth parts
		(vi)	Flame cells

(1) a-(v), b-(iii),(i) c-(ii), d-(iv), e-(vi)

(2) a-(iii), b-(ii), c-(v), d-(iv), e-(vi)

(3) a-(v), b-(ii), (c)-(iii), d-(iv), (e)-(vi)

(4) a-(v), b-(iii), c-(ii), d-(vi), e-(iv)

27) Read the statements (A–E) and choose the option which include incorrect one :-
(A) Parapodia are lateral appendages in arthropods used for swimming
(B) In mollusca, radula is an excretory organ
(C) Aschelminthes are dioecious
(D) Adult echinoderms show radial symmetry
(E) Ctenophorans are diploblastic

(1) A and C

(2) A and B

(3) C and E

(4) A, C and D

28) Which of the following pairs are correctly matched?
(A) *Poikilotherms* – Birds and mammals
(B) *Agnatha* – Petromyzon and Myxine
(C) *Tunicata* – Ascidia, Salpa and Doliolum

(D) *Cartilaginous fishes* – Betta and Labeo

(1) A and B

(2) A, B and C

(3) B and C

(4) A, B and D

29) Endoskeleton of calcareous ossicles, external fertilization, adults show radial symmetry, development indirect.
The above-given feature is true for:

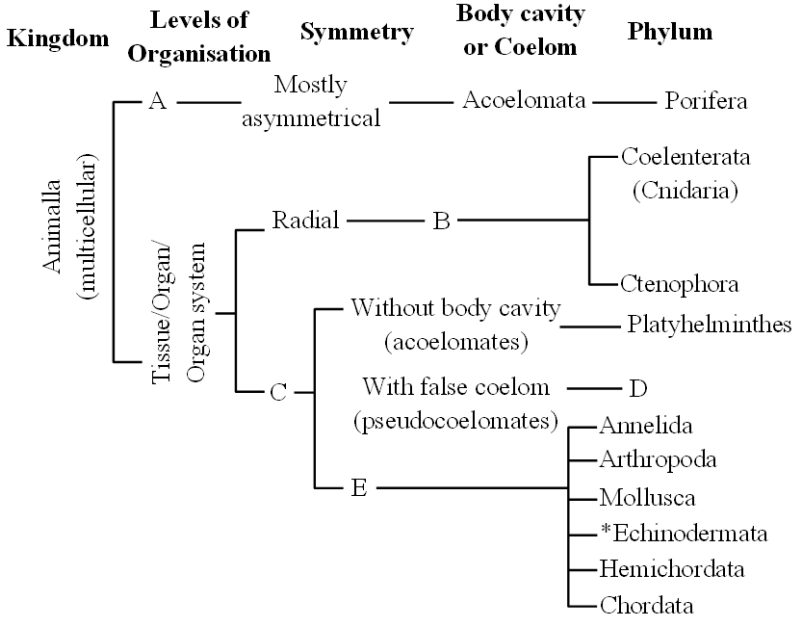
(1) Echinus, Asterias

(2) Pila, Ophiura

(3) Antedon, Aplysia

(4) Pinctada, Cucumaria

30) Study the flow chart given below and identify the missing parts A, B, C, D, E.



	A	B	C	D	E
(1)	Cellular	Acoelomata	Bilateral	Aschelminthes	Coelomates
(2)	Cellular	Coelomata	Radial	Aves	Pseudocoelomates
(3)	Cellular	Acoelomata	Radial	Mammalia	Pseudocoelomates
(4)	Cellular	Coelomata	Radial	Aschelminthes	Coelomates

(1) 1

(2) 2

(3) 3

(4) 4

31) Consider the following statements
During the alternation of generation or metagenesis in dimorphic cnidarians the polyps are :-

(A) Sessile
(B) Terrestrial
(C) Polyp forms the medusae sexually
(D) Polyp forms the medusae asexually

(1) A, C & D are correct
(2) A and D are correct
(3) B, C & D are correct
(4) A, B, C are correct

32) Which of the following is incorrect match :-

(1) Economically important insects - *Apis, Bombyx, Laccifer*
(2) Vectors - *Anopheles*, *Culex*, *Aedes*
(3) *Pinctada* - Silk
(4) Living fossil - Limulus

33) Read the following four statements A-D :-
(A) All schizocoelomate are protostomes.
(B) All protostomes are schizocoelomate.
(C) All schizocoelomate animals are triploblastic.
(D) All bisexual animals show self fertilization.
How many of the above statements are incorrect?

(1) One
(2) Four
(3) Two
(4) Three

34) Match the following animals with their respective characteristics :-

(a)	Unio	(i)	Jointed appendages
(b)	Nereis	(ii)	Bioluminiscence
(c)	Ctenoplana	(iii)	Mantle
(d)	Locust	(iv)	Parapodia

	(a)	(b)	(c)	(d)
(1)	(iii)	(ii)	(i)	(iv)
(2)	(iii)	(iv)	(ii)	(i)
(3)	(ii)	(iv)	(iii)	(i)
(4)	(iii)	(ii)	(iv)	(i)

(1) 1
(2) 2
(3) 3
(4) 4

35) Which of the following is not an insect :-

(1) Honey bee
(2) Butterfly
(3) Scorpion
(4) Mosquito

36) Closed circulatory system first appeared in which phylum:-

(1) Arthropoda
(2) Mollusca
(3) Annelida
(4) Chordata

37) Which of the following is an ectoparasite ?

(1) *Petromyzon*
(2) *Rana*
(3) *Pheretima*
(4) *Taenia*

38) Coelom is lined by

(1) Ectoderm
(2) Mesoderm
(3) Endoderm
(4) Mesogloea

39) **Statocyst** is related with :-

(1) Excretion
(2) Respiration
(3) Balance
(4) Hearing

40) Respiration in Delphinus takes place by :-

(1) Gills
(2) Skin
(3) Trachea
(4) Lungs

41) Dengue fever is transmitted by which :

(1) Anopheles mosquito
(2) Aedes mosquito
(3) Culex mosquito
(4) Tse-Tse fly

42)

Persistent notocord is found in :-

(1) *Exocoetus*
(2) *Betta*
(3) *Scoliodon*
(4) *Labeo*

43) Retrogressive metamorphosis is not found in:-

(1) *Ascidia*
(2) *Herdmania*
(3) *Salpa*
(4) *Balanoglossus*

44) Which one of the following statement about all the four of Sycon, Nereis, Dolphin and Pavo is correct?

(1) All are bilaterally symmetrical
(2) *Pavo* is homeothermic while the remaining three are poikilothermic
(3) *Nereis* is a fresh water form while all others are marine
(4) Sycon has special collared cells called choanocytes, not found in the remaining three.

45) Which is the chordate animal who have only single fundamental chordate character during their adult stage while remaining fundamental features disappear during metamorphosis?

(1) *Amphioxus*
(2) *Balanoglossus*
(3) *Petromyzon*
(4) *Ascidia*

46) **Statement-I :** Statocyst is balancing organ present in Arthropoda.
Statement-II : Radula is a file like rasping organ present in Echinodermata.

(1) Both Statement I and Statement II are incorrect.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct
(4) Both Statement I and Statement II are correct.

47) Identify the fresh water bony fishes.

(1) Flying fish, Catla
(2) Magur, Sea horse
(3) Rohu, *Clarias*
(4) Saw fish, *Labeo*

48) See following specimens and give correct answer about them :-





How many of the following features are not common in both ?

(A) Marine (B) Air bladder
(C) Cold blooded (D) Terminal mouth
E) Oviparous (F) Presence of scales

(1) Two
(2) Three
(3) Four
(4) Five

49) **Assertion :** Chondrichthyes have internal fertilisation.
Reason : Male bears claspers, which help in transfer of sperms in female fish.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

50) **Assertion :-** Annelida, Arthropoda , Mollusca and Chordates are metamerically segmented animals.
Reason :- Body is externally and internally divided into segments with serial repetition of at least some organs.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Assertion is false, but Reason is true.

51) Different cells have different sizes. Arrange the following cells in an ascending order of their size.
Choose the correct option among the following.
I. *Mycoplasma*
II. Ostrich eggs
III. Human RBC
IV. Bacteria

(1) I, IV, III, II
(2) I, II, III, IV
(3) II, I III, IV

(4) III, II, I, IV

52) Arrangement of microtubules in flagella is–

(1) 9+2
(2) 0+9
(3) 11+0
(4) 12+0

53) Ribosomes are made up of–

(1) DNA + Protein
(2) RNA + Protein
(3) DNA + RNA
(4) Only protein

54) Which one of the following structure is called "Organelle within organelle"?

(1) ER
(2) Ribosome
(3) Mitochondria
(4) Chloroplast

55) Kinetochore is present on–

(1) Mitochondria
(2) Cell wall
(3) Flagella
(4) Chromosome

56) Golgi body is concerned with–

(1) Respiration
(2) Photosynthesis
(3) Secretion
(4) Digestion

57) Cisternae are related with–

(1) Flagella
(2) Chloroplast
(3) ER and Golgibody
(4) Ribosomes

58) Main site of aerobic respiration is–

(1) Ribosome
(2) Golgibody
(3) Mitochondria
(4) Chloroplast

59) Which maintains cell shape and associated with cellular movements and intracellular transport ?

(1) Cytoplasm
(2) Golgibody
(3) Cytoskeleton
(4) Mitochondria

60) Oxidative phosphorylation is related with :-

(1) Chloroplast
(2) Mitochondria
(3) Lysosome
(4) GERL system

61) The content of the nucleolus is continuous with the rest of the ……..as it is a …… structure.

(1) Nucleoplasm, Double membranous
(2) Nucleoplasm, Membrane less
(3) Cytoplasm, Membrane less
(4) Cytoplasm, Single membranous

62) Endomembrane system include :-

(1) Ribosome, Golgi bodies, ER, Lysosome
(2) Vacuole, Lysosome, Golgi bodies, ER
(3) ER, Golgi bodies, Lysosome, Peroxisome
(4) Lysosome, ER, Ribosome, Golgi bodies

63) Formation of steroidal hormones is a specific function of :-

(1) RER
(2) Golgi body
(3) Ribosome
(4) SER

64) Number of mitochondria in a cell depends upon

(1) shape of that particular cell
(2) size of that particular cell
(3) colour of that particular cell
(4) physiological activity of that particular cell

65) i. Form the basal body of cilia and flagella
ii. Play important role in animal cell division
iii. Cart wheel like structure
iv. Have central hub and 9 peripheral triplet fibrils
Above given character's associated with -

(1) Cillia
(2) Flagella
(3) Centriole
(4) Peroxisome

66) Name chromatin was proposed by :-

(1) Robert Brown
(2) Flemming
(3) Robert Hook
(4) Leeuvenhoke

67) Which of following structure contain ribosome.

(1) Outer membrane of nuclei
(2) Inner membrane of nuclei
(3) Cytoplasm and RER
(4) More than one option are right

68) Membrane bound minute vesicles contain various enzymes are.

(1) Microbodies
(2) Ribosome
(3) Centriole
(4) Centrosome

69) The core part of the cilia and flagella are known as__A__is covered with__B__.

(1) A–Axoneme; B–Cell membrane
(2) A–Cell membrane; B–Axoneme
(3) A–Axoneme; B–Cell wall
(4) A–Axoneme; B–Actin

70) Type of chromosome which has terminally located centromere :-

(1) Metacentric chromosome
(2) Sub-meta centric chromosome
(3) Telocentric chromosome
(4) Acrocentric chromosome

71) Match the column

	Column-I		Column-II
a.	Typical eukaryotic cell	i.	7.0 µM
b.	Viruses	ii.	0.1 µM
c.	PPLO	iii.	0.02–0.2 µM
d.	Typical bacterium	iv.	1–2 µM
e.	Human RBC	v.	10–20 µM

(1) a–ii; b–iii; c–v; d–iv; e–i

(2) a–iv; b–ii; c–ii; d–v; e–i

(3) a–v; b–iii; c–ii; d–iv; e–i

(4) a–i; b–iii; c–ii; d–v; e–iv

72) Match the Column:

	Column-I		Column-II
(A)	Chromatin	(i)	Excretion & Osmoregulation
(B)	Nucleolus	(ii)	Non staining part located at constant location of chromosome
(C)	Secondary Constriction	(iii)	Nucleoprotein fibre's
(D)	Contractile vacuole	(iv)	r-RNA synthesis

(1) (A) - (iii), (B) - (iv), (C) - (i), (D) - (ii)

(2) (A) - (iii), (B) - (ii), (C) - (iv), (D) - (i)

(3) (A) - (iii), (B) - (iv), (C) - (ii), (D) - (i)

(4) (A) - (i), (B) - (iv), (C) - (ii), (D) - (iii)

73) Interphase subdivided in to

(1) M, G, S, G_2

(2) G_1, S, G_2

(3) G_1, G_0, G_2, S

(4) S, G_2, M

74) **Assertion :** The tonoplast faciliates the transport of a number of ions and other materials against concentration gradient into the vacuole.
Reason : The concentration of ions and other materials is significantly lower in the vacuole than in

the cytoplasm.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

75) **Statement-I** : The interphase nucleus has highly extended and elaborate nucleoprotein fibres called chromatin.
Statement-II : During cell division, cells show structured chromosomes in place of the nucleus.

(1) Statement-I & Statement-II both are correct.
(2) Both Statement-I & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Statement-I is correct & Statement-II is incorrect.

76) Which of the following cell wall is capable in growth ?

(1) Secondary cell wall
(2) Primary cell wall
(3) Middle lamella
(4) Tertiary cell wall

77) How many organelle of the cell are double membranous involved in photorespiration ?

(1) One
(2) Two
(3) Three
(4) Four

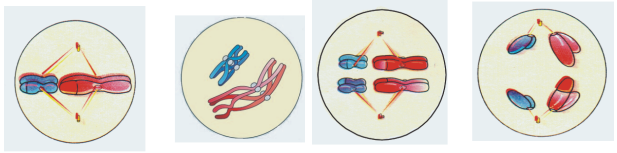
78) In oocytes of some vertebrates which stage can last for months or years-

(1) Leptotene
(2) Pachytene
(3) Diplotene
(4) Diakinesis

79) Choose the correct statement.

(1) Physico-chemical approach to study and understand living organisms is called reductionist biology.
(2) Cells of all living organisms have a nucleus.
(3) All cell are formed de novo from abiotic materials.
(4) Lysosomes and ribosome are endomembrane structures.

80) Arrange the following events of meiotic cell division diagrams in correct order :-


(A) (B) (C) (D)

(1) A → C → D → B

(2) B → C → D → A

(3) C → A → B → D

(4) B → A → C → D

81) Arrange thefollowing events of meiosis in the correct sequence -
A. Terminalisation of chiasmata
B. Crossing over
C. Synapsis
D. Disjunction of chromosomes
E. Dissolution of synaptonemal complex
The correct sequence is-

(1) A → B → C → D → E

(2) E → D → C → B → A

(3) C → A → D → E → B

(4) C → B → E → A → D

82) Consider the following four statements (I-IV) related to cell cycle, and select the correct option stating them as true [T] and false [F] -
I. Cell growth (in terms of cytoplasmic increase) is a continuous process.
II. Every chromosome essentially has a secondary constriction.
III. The number of chromosomes doubles in S-phase.
IV. The cells that do not divide further exit. G1- phase to enter an inactive stage.

(1) I → T; II → F; III → F; IV → F

(2) I → F; II → T; III → T; IV → T

(3) I → F; II → F; III → T; IV → T

(4) I → T; II → F; III → F; IV → T

83) Which of the following statements is wrong w.r.t cell cycle-
(a) M-phase lasts more than 95% duration for cell cycle.
(b) In human cell, the average duration of cell cycle is 24 hour
(c) Cells are metabolically active during G_0 stage
(d) Chromosome number becomes double in S-phase

(1) Only (a)

(2) (a) and (b)

(3) (b) and (c)

(4) (a) and (d)

84) In meiosis how many cycle of DNA replication occur?

(1) One

(2) Two

(3) Three

(4) No replication occur

85) Match List-I with List-II

	List-I		**List-II**
(A)	Centrosome	(I)	Primary constriction in chromosome
(B)	Thylakoid	(II)	Infolding in mitochondria
(C)	Cristae	(III)	Cart wheel structure
(D)	Centromere	(IV)	Flattened membranous sac in the stroma of chloroplast

(1) A-III, B-IV, C-II, D-I

(2) A-I, B-III, C-II, D-IV

(3) A-III, B-II, C-I, D-IV

(4) A-I, B-IV, C-II, D-III

86) Each cell of Onion root tip has 16 chromosomes. How many chromosomes will the cell have at G_1 phase, after s phase and after M phase?

(1) 8, 16, 32

(2) 16,16, 32

(3) 16, 16, 16

(4) 16, 32, 8

87) Match List-I with List-II

	List-I (Stages of Mitosis)		**List-II (Events)**
A	Prophase	(I)	Chromatids move to opposite poles
B	Metaphase	(II)	Spindle fibres attach to kinetochore of chromosomes
C	Anaphase	(III)	Nucleolus, Golgi complex and ER reform
D	Telophase	(IV)	Chromatin condensation

(1) A-II, B-IV, C-I, D-III

(2) A-IV, B-II, C-I, D-III

(3) A-III, B-II, C-I, D-IV

(4) A-III, B-IV, C-II, D-I

88) How many statement are correct
(i) In diakinesis the chromosomes are fully condensed.
(ii) Diakinesis represent transition to metaphase.
(iii) There in no replication of DNA during interkinesis.

(iv) At the end of telophase-I althaough in many cases the chromosomes do undergo some dispersion, they do not reach the extremely extended state of interphase nucleus.

(1) 3

(2) 1

(3) 4

(4) 2

89) Which of the the statement is not correct

(1) Although cell growth in terms of cytoplasmic increase is a continuous process, DNA synthesis occurs only during one specific stage in the cell cycle

(2) Cells in quiescent stage of cell cycle remains metabolically active but not longer proliferate unless called on to do so depending on the requirement of the organism

(3) Plants show mitotic divisions in diploid cell only.

(4) Normal mitosis can not occur without DNA replication in the S phase.

90)

	Column-I		**Column-II**
A	Homologous chromosomes separating	1	Anaphase-I
B	Sister chromatids separating	2	Anaphase-II
C	Chiasmata	3	Diplotene
D	Reduction in chromosome number	4	Meiosis-I

(1) A-4, B-2, C-3, D-1

(2) A-2, B-1, C-3, D-4

(3) A-1, B-2, C-3, D-4

(4) A-3, B-1, C-2, D-4

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	2	1	4	4	1	1	2	1	4	1	3	2	2	4	1	3	1	2	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	1	4	2	3	3	3	1	4	1	1	1	2	2	2	2	2	1	3	3
Q.	41	42	43	44	45															
A.	4	1	2	3	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	2	3	1	4	2	4	4	3	2	1	3	1	1	4	4	3	3	3	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	3	4	4	4	1	2	3	2	4	2	1	3	4	3	4	4	1	4	3
Q.	86	87	88	89	90															
A.	2	1	3	3	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	1	3	2	3	3	3	1	1	2	4	1	3	2	3	2	2	3	4	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	3	1	1	3	1	2	3	1	1	2	3	3	2	3	3	1	2	3	4
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	3	4	4	4	2	3	2	1	4	1	1	2	2	4	3	3	3	3	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	2	4	4	3	2	4	1	1	3	3	3	2	3	1	2	2	3	1	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	4	4	1	1	3	2	3	3	3										

SOLUTIONS

PHYSICS

1)

5 sin (100° + 27°)
= 5 sin (90° + 37°)
= 5 cos (37°) = 4

2)

cos (106°) = cos (2 × 53°)
$= 2\cos^2(53°) - 1$
$= \frac{18}{25} - 1$
$= \left(\frac{-7}{25}\right)$

3)

$3x - 2y + 4 = 0 \Rightarrow 2y = 3x + 4$
$y = \frac{3}{2}x + \frac{4}{2} \Rightarrow y = +\frac{3}{2}x + 2$

slope = $+m = \frac{3}{2}$ & c = +2

4) $m = \frac{2-0}{0-3} = \left(-\frac{2}{3}\right)$

5) $\frac{d}{dx}(x^5 + x^4 + 7) = 5x^4 + 4x^3$

6) First derivative: $\frac{d}{dx}(x^n) = nx^{n-1}$
Second Derivative: $\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$
Solution :
$y = 6x^3 - 4x$
$\Rightarrow \frac{dy}{dx} = 18x^2 - 4$
$\frac{d^2y}{dx^2} = 36x$

at x = 1 $\frac{d^2y}{dx^2}$ = 36

7) $y = x^2 + 4x$

$\frac{dy}{dx} = 2x + 4$

$\frac{dy}{dx} = 0$

$\Rightarrow 2x + 4 = 0$

$\Rightarrow 2x = -4$

$x = -2$

$\frac{d^2y}{dx^2} = 2 > 0$

$\therefore$ x = –2 is a point of minima

$y_{(x = -2)} = (-2)^2 + 4\times (-2) = 4 - 8 = -4$

8) $\frac{dA}{dt} = (10t + 4)$

At (t = 3)

$\frac{dA}{dt} = 10(3) + 4$

$= 34\ m^2/s$

9) $\int 4\sin x dx - \int \frac{2}{x} dx$

= –4 cos x – 2ln x + c

10)

The average value of function f(x) in the interval [a, b] is given by

$f_{avg} = \frac{1}{b-a}\int_a^b f(x)dx$

Given here, y = f(x) = 2x + 3, a = 0, b = 1

$y_{avg} = \frac{1}{1-0}\int_0^1 (2x+3)dx$

$y_{avg} = \frac{\left(x^2+3x\right)_0^1}{1-0}$

$y_{avg} = \left(1^2 + 3\times 1\right) - (0) = 4$

11) **Solution :**

The roots α and β are given by

$\alpha \text{ and } \beta = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

Given $2x^2 - x - 3 = 0$

Comparing with $ax^2 + bx + c$

a = 2, b = -1, c = -3

α and $\beta = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(2)(-3)}}{2 \times 2}$

$= \frac{1 \pm \sqrt{1+24}}{4} \Rightarrow \frac{1 \pm 5}{4}$

$\Rightarrow \frac{1+5}{4} \text{or} \frac{1-5}{4}$

$\Rightarrow \frac{6}{4} \text{or} -\frac{4}{4} = \frac{3}{2}$ **or** -1

Final Answer : option (1)

12) $S = \frac{16}{2}(4 + 64)$

$= 8 \times 68$

$= 544$

13)

By logarithmic properties

$\log (m \times n) = \log m + \log n$ and $\log (m^n) = n \log m$

$\log 18 = \log\left(2 \times 3^2\right)$

$\log 18 = \log 2 + \log\left(3^2\right)$

$\log 18 = \log 2 + 2\log 3$

$\log 2 = 0.3$

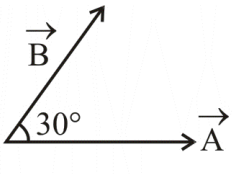
$\log 3 = 0.5$

$\log 18 = 0.3 + 2 \times 0.5$

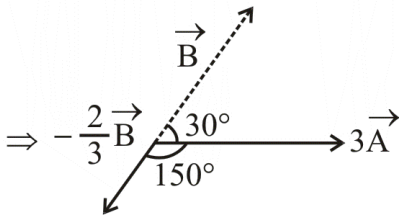
$\log 18 = 0.3 + 1.0 = 1.3$

Final answer: Option (2)

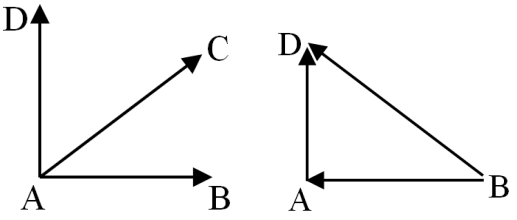
14)



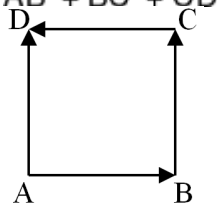
$\Rightarrow -\frac{2}{3}\vec{B}$



15) $\vec{AB} + \vec{AD} = \vec{AC}$ $\vec{BA} + \vec{AD} = \vec{BD}$



$\vec{AB} + \vec{BC} + \vec{CD} = \vec{AD}$



16)

Given: $F_1 = F_2 = F$, $R = 40\sqrt{3}N$ and $\theta = 60°$

$R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}$

$\Rightarrow 40\sqrt{3} = \sqrt{F^2 + F^2 + 2F^2\cos 60} = \sqrt{3F^2}$

$\Rightarrow$ F = 40N

17) Here, $\vec{A} + \vec{B} = \vec{C}$ and $\vec{B} = \vec{C} - \vec{A}$

Since $\vec{C} \perp \vec{A}$ therefore

$B^2 = C^2 + A^2$

Also $|\vec{C}| = |\vec{A}|$ therefore

$B^2 = 2A^2 \Rightarrow B = \sqrt{2}A$

Now, $A^2 + B^2 + 2AB\cos\theta = C^2 = A^2$

Therefore we find $\cos\theta = -\frac{1}{\sqrt{2}}$

this gives $\theta = \frac{3\pi}{4}$

18)

$2A\sin\left(\frac{\theta}{2}\right) = A$

$\sin\left(\frac{\theta}{2}\right) = \frac{1}{2}$

$\theta = 60°$

19) $\left|\hat{a} - \hat{b}\right| = 2\,|\hat{a}|\sin\left(\frac{60°}{2}\right) = 2\left(\frac{1}{2}\right) = 1$

20) Sum of given vectors : $\left(\hat{i} - 2\hat{j} + 2\hat{k}\right) + \left(2\hat{i} + \hat{j} - \hat{k}\right) = 3\hat{i} - \hat{j} + \hat{k}$

Let $\vec{P}$ be the vector to be added to the sum of given vectors :

$\therefore\ 3\hat{i} - \hat{j} + \hat{k} + \vec{P} = \hat{i}$

$\vec{P} = \hat{i} - (3\hat{i} - \hat{j} + \hat{k}) = -2\hat{i} + \hat{j} - \hat{k}$

21) $\left(\vec{A}+\vec{B}\right)$ is perpendicular to $\left(\vec{A}-\vec{B}\right)$

Thus $\left(\vec{A}+\vec{B}\right) \cdot \left(\vec{A}-\vec{B}\right)=0$

$\Rightarrow A^2+\vec{B}\cdot\vec{A}-\vec{A}\cdot\vec{B}-B^2=0$

$\Rightarrow A^2 - B^2 = 0 \Rightarrow A = B$

Thus the ratio of magnitudes A/B = 1

22) $\cos\alpha = \frac{2}{\sqrt{45}}$, $\cos\beta = \frac{4}{\sqrt{45}}$; $\cos\gamma = \frac{-5}{\sqrt{45}}$

23) $\cos(\theta) = \frac{42+24-12}{\left(\sqrt{81}\right)\sqrt{81}}$

$\cos\theta = \frac{54}{81} = \frac{2}{3}$

$\sin\theta = \sqrt{1-\left(\frac{2}{3}\right)^2} = \left(\frac{\sqrt{5}}{3}\right)$

24) **Question Explanation :**

Find the component of $\vec{A}=\hat{i}+2\hat{j}+2\hat{k}$ on $\vec{B}=\hat{j}+\hat{k}$:

Concept :

Application of dot product.

Solution :

Component of $\vec{A}$ on $\vec{B}$ = A cos θ . $\hat{B}$

$= A \times \frac{\vec{A}.\vec{B}}{AB}.\hat{B}$

$= \frac{\vec{A}.\vec{B}}{B}.\hat{B} = \frac{(\vec{A}.\vec{B}).\vec{B}}{B^2}$

$= \frac{(2+2)}{\left(\sqrt{1+1}\right)^2} \times (\hat{j}+\hat{k})$

$= 2\hat{j}+2\hat{k}$

Final Answer :

Option (2)

25) $\hat{n} = \frac{\vec{A}\times\vec{B}}{|\vec{A}\times\vec{B}|}$

26) $\vec{d}_1 = 30\hat{j}$

$\vec{d}_2 = 20\hat{i}$;

$\vec{d}_3 = -30\hat{i}-30\hat{j}$

$\vec{d}_{net} = (-10\hat{i})\ m$
$\vec{d}_{net} = 10\ m$ along west

27) $|\vec{v}_{avg}| = \frac{80}{40} = 2$ m/s

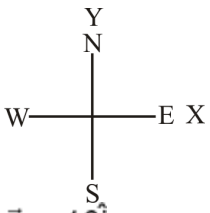
28) $v_{avg} = \frac{2(10)(15)}{25}$
$= 12$ m/s

29) $\Delta\vec{v} = (-10\hat{j}) - (10\hat{i})$
$= \left(10\sqrt{2}\right)$ m/s S-W
$= 14.14$ m/s S-W

30)

Question Explanation : Body is moving with velocity 10 m/s towards north and after 2 sec. Its velocity becomes 20 m/s towards west. Find average acceleration of body.

Concept: Kinematics - average acceleration.
Formula : $\vec{a}_{av} = \frac{\vec{v}_f - \vec{v}_i}{\Delta t}$
Solution:



$\vec{v}_i = 10\hat{j}$
$\vec{v}_f = -20\hat{i}$
$\vec{a}_{av} = \frac{\vec{v}_f - \vec{v}_i}{\Delta t}$
$= \frac{-20\hat{i} - 10\hat{j}}{2}$
$|\vec{a}_{av}| = \frac{\sqrt{20^2 + 10^2}}{2}$
$= 5\sqrt{5}\ m/s^2$

Final answer : option (1)

31)

$v = \frac{dx}{dt}$
$v = 2at$
$\frac{dv}{dt} = (2a)$

32)

$v = \frac{dx}{dt} = 6t^2 - 6t + 4$ .....(1)

$a = \frac{dv}{dt} = 12t - 6$ .....(2)

$0 = 12t - 6$

$t = \left(\frac{1}{2}\right)$ sec.

$v = 6\left(\frac{1}{2}\right)^2 - 6\left(\frac{1}{2}\right) + 4$

$= 1.5 - 3 + 4$

$= 2.5$ m/s

33) **Explain question :** Velocity depends on the position, then acceleration at the given position will be.

Concept: This question is based on motion with variable acceleration.

Solution :

$v = 3x^2 - 2x$

$\frac{vdv}{dx} = a = \left(3x^2 - 2x\right)(6x - 2)$

$a(x = 2) = (3(2)^2 - 2 \times 2)(6 \times 2 - 2)$

$= (3 \times 4 - 4)(12 - 2)$

$= 8 \times 10 = 80$ m/s^2

Final answer : 80 m/s^2

34) Area under a-t graph = change in velocity

$\frac{1}{2} \times 5 \times 6$ = v – u

v = u + 15

u = 10 m/s

Hence,

$V_{max} = 10 + 15 = 25$ m/s

35)

Velocity of car 'P' = $v_P = \frac{dx_P}{dt} = a + 2bt$

Velocity of car 'Q' = $v_Q = \frac{dx_Q}{dt} = f - 2t$

Velocity of cars are same ; $v_P = v_Q$

⇒ a + 2bt = f – 2t

⇒ 2t + 2bt = f – a

⇒ $t = \frac{f - a}{2(b + 1)}$

36) $\Delta v = \int_0^t (at)dt$

$v - u = \frac{at^2}{2}$

$v = u + \frac{at^2}{2}$

37) $\vec{v} = \vec{u} + \vec{a}\,t$

$\vec{v} = (4\hat{i} + 3\hat{j}) + (0.3\hat{i} + 0.4\hat{j})10$

$\vec{v} = (7\hat{i} + 7\hat{j}) \Rightarrow |\vec{v}| = 7\sqrt{2}$ m/s

38) $S_{5^{th}} = 10 - \left(\frac{2}{2}\right)(2 \times 5 - 1)$

= 10 – 9

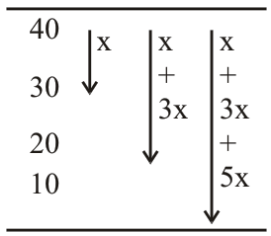
= 1 m

39) As Given $S_{n^{th}} = S_5$ [Also u = 0 & a = g]

$\Rightarrow \frac{g}{2}(2n-1) = \frac{1}{2}g(5)^2$

$\Rightarrow$ n = 13 sec

40)



x + 3x + 5x = 99

9x = 99

x = 11m

height = (9x – x)

= 8x

= 88 m

41)

h_{max} = Area of (v – t) [0 to 120]

$= \frac{1}{2}(120) \times (1000)$

= 60 km

42) Slope of v-t curve is acceleration on which is (+) for t = 0 to t = 2 sec. and (–) for t = 2 sec. to t = 4 sec.

43) If two projectiles are projected at complementary angles i.e. θ and 90° – θ with the same speed, they cover equal range.
For projectile - Case-1
θ_1 = 60° (with horizontal)
For projectile - Case-2
θ_2 = 90° – 60° = 30° (with horizontal)
Since both angles are complementary and speed is same in both cases, range is same in both cases.
i.e. $R_1 = R_2$ = 100 M
Note : You can also solve with the help of formula i.e. R = $u^2\frac{\sin 2\theta}{g}$

44) In horizontal projectile motion x-ut for vertical fall, $y = \frac{1}{2}gt^2$ (by equation $S_y = 4yt + \frac{1}{2}ayt^2$)
For safe landing vertical fall < difference in height i.e. 25 – 5 = 20 m
We need to calculate is required to ensure horizontal distance = 8m between building before falling 20m
Time taken to fall 20 m

$$T = \sqrt{\frac{2H}{g}} = \sqrt{\frac{2 \times 20}{10}} = 2 \text{ sec}$$

To cover horizontal distance 8m in this time $\mu = \frac{x}{t} = \frac{8}{2} = 4$ m/sec.

45)

$$t = \frac{\text{distance}}{\text{speed}} = \frac{120 + 130}{20 + 30} = \frac{250}{50} = 5 \text{ sec.}$$

CHEMISTRY

46) **Explain Question:** Check correctness of assertion–reason about electron probability
Concept: Schrodinger's wave equation
Solution: $|\Psi|^2$ gives probability of finding electron in a region; quantum mechanics cannot determine exact position of an electron, only probability
Final Answer: Option (3)

47) St.-1 :- $E = -13.6 \times \frac{z^2}{n^2}$
For H-atom in n = 1 ; E = –13.6 eV
For He^+ in n = 2, $E = -13.6 \times \frac{(2)^2}{(2)^2}$
= –13.6 eV
St.1 is correct
St.-2 :- Incorrect (uncertainty principle)

48)

Explanation : Match counts of electrons (with given quantum numbers / shells / kernel) for a copper atom (Z = 29) to the numbers in Column-II.
Concept : Electron configuration and shell occupancy
Solution : Z = 29 → electronic configuration: [Ar] $3d^{10}$ $4s^1$.
(a) n = 4, □ = 0, m = 0 → orbital 4s contains 1 electron → 1.
(b) n = 3, s = +1/2 → n = 3 has $3s^2$, $3p^6$, $3d^{10}$ = 18 electrons; half with s = +1/2 → 9.
(c) Penultimate shell = n = 3 → 18 electrons.
(d) Kernel = all electrons except outermost shell (n = 4) → 29 − 1 = 28.
Final Answer : Option (3)

49) **Explain the Question :** Determine the number of subshells and electrons for principal quantum number n = 4.
Concept : This question is based on number of subshells = n, electrons per shell = $2n^2$.
Solution :
Number of subshells : n = 4 ⇒ 4 subshells → 4s, 4p, 4d, 4f
Maximum electrons in n = 4 shell.
$2n^2 = 2 \times 4^2 = 32$
Final Answer : 1

50)

e > p > α > n

51)

Explain Question : We need to calculate % change in atomic weight of C^{12} if neutron mass is doubled, proton mass is halved, and electron mass is doubled.

Concept : Atomic weight ≈ total mass of protons + neutrons (electron mass negligible).

Solution :

C^{12} contains 6proton and 6neutron
Initial atomic mass : 6p + 6n = 12amu
New atomic mass : Protons → $6 \times \left(\frac{1}{2}\right)$amu = 3amu , Neutrons → 6 × (2amu) = 12amu.
Total final atomic mass = 15amu
% Change = $\left[\left(\frac{15-12}{12}\right)\right] \times 100 \left(\frac{3}{12}\right) \times 100 = 25\%$

Final Answer : Option (2)

52)

Question Explanation : To find the kinetic and potential energy of an electron in the second Bohr orbit (n=2) of hydrogen (Z=1).

Concept : This question is based on energy of electron in n^{th} orbit.

Solution :

Total energy $E_n = -\frac{13.6Z^2}{n^2}eV$

Kinetic energy $KE = -E_n$.

Potential energy $PE = 2E_n$.

$E_2 = -\frac{13.6 \times 1^2}{2^2} = -\frac{13.6}{4} = -3.4\text{eV}.$

$KE = -(-3.4) = +3.4\text{eV}.$

$PE = 2 \times (-3.4) = -6.8\text{eV}$

Final Answer : option (4)

53) **Question Explanation :**
We are to identify which given expression related to Bohr's atomic model are correct.

Concept :
Bohr's atomic model equations for energy and orbit radius.

Solution :
$E_n = -2.18 \times 10^{-18} \times (Z^2/n^2)$ → Correct energy formula for n^{th} orbit.
$\Delta E = 2.18 \times 10^{-18} \times Z^2 (1/n_1^2 - 1/n_2^2)$ → Correct energy difference expression.
$rn = 52.9 \times (n^2/Z)$ pm ® Correct radius formula for hydrogen - like atoms.
All three are standard and correct.

Final Answer :
Option (4)

54) d_{zx}
↓
Nodel plane (xy and yz)

55) $x = \frac{n_1^2}{R_2^2} = \frac{1}{R},$
for Balmer $n_1 = 2$, n2 = 3
$\lambda = \frac{1}{R}\left[\frac{4 \times 9}{9-4}\right] = \frac{36}{5}$

56) **Explain the question:** Classify atomic sets based on particle counts.
Concept: Nuclidic types: isotopes (Z), isobars (A), isotones (N), isodiaphers (A-2Z).
Solution:
In set (i), the elements share an identical atomic number Z = 26, defining them as **isotopes**.
In set (ii), both atoms possess a mass number A = 3, identifying them as **isobars**.
For set (iii), we calculate the number of neutrons using N = A – Z, yielding 76 – 32 = 44 and 77 – 33 = 44; thus, they are isotones.

In set (iv), the difference between neutrons and protons (A – 2Z) is consistent: 235 – 184 = 51 and 231 – 180 = 51, classifying them as isodiaphers.
Finally, set (v) includes species with the same atomic number Z = 1, designating them as isotopes.
Following these specific chemical relationships leads directly to the combination presented in Option 1.
Correct Order is: [(i) - a], [(ii) - d], [(iii) - b], [(iv) - c], [(v) - a]
Final Answer: (Option 1)

57) $\lambda = \frac{h}{P}; \frac{\lambda_A = \frac{h}{P_A}}{\lambda_B = \frac{h}{P_B}} = \frac{P_B}{P_A}$

$\therefore \lambda_{B} = \frac{\lambda_A \times P_A}{P_B} = \frac{5 \times 10^{-8} \times P_A}{P_A/2}$

$\therefore \lambda_B = 10^{-7}$ m.

58) $\nu \propto \frac{Z^2}{n^3}$

$\frac{\nu_{He^+}}{\nu_{H^+}} = \frac{(2)^2/(3)^3}{(1)^2/(2)^3}$

$\Rightarrow \frac{8 \times 4}{27} = \frac{32}{27}$

59) Given : Power of bulb (P) = 100 W and wavelength (λ) = 400 nm = 400 × 10^{-9} m. We know that energy of one photon

(E) = $\frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34}) \times (3 \times 10^8)}{400 \times 10^{-9}}$

= 5 × 10^{-19} J.

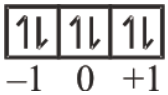
Therefore no. of photons emitted per second (n) = $\frac{P}{E} = \frac{100}{5 \times 10^{-19}} = 2.0 \times 10^{20}$

(where c=velocity of light equal to 3×10^8 ms^{-1} and h=Planck's constant equal to 6.63×10^{-34} Js)

60)

(a)

Mg =	$1s^2$	$2s^2$	$2p^6$	$3s^2$	
ℓ =	0	0	1	0	
m =	0	0	1	0	for m = 0
					No. of e^- = 8



(b) One orbital occupied maximum $2e^-$

(c) M shell n = 3, no. of e^- = $2n^2 = 2 \times 3^2 = 18$

(d) P = $1s^2 2s^2 2p^6 3s^2 3p^3$

l = 0 for s orbital, so $6e^-$, in s, so for 3e, s = $+\frac{1}{2}$

and for 3e, s = $-\frac{1}{2}$

61)

Vapour density is unit less quantity.
V_m at STP = 22.4 L

62) No. of atoms = $\frac{16}{32} \times 2N_A = N_A$
No. of atoms = $\frac{16}{48} \times 3N_A = N_A$.

63) **Explain Question:** We need to match given amounts of different gases to their **volume at STP** (1 mole gas =
22.4 L at STP).
Concept: This question is based on Moles = mass/Molar mass, Number of atoms=Moles × NA, Moles=Volume of gas in L / 22.4 L/mol
Solution:
(A) 14 g of N_2 → Molar mass N_2 = 28 g → 14 g = 0.5 mol → Volume = 0.5 × 22.4 = **11.2 L (R)**
(B) 6.02×10^{22} molecules NH_3 → 1 mol = 6.022×10^{23} → given = 0.1 mol → Volume = 0.1 × 22.4 = **2.24 L (S)**
(C) 1 mol SO_2 → Volume = **22.4 L (P)**
(D) 0.2 mol O_3 → Volume = 0.2 × 22.4 = **4.48 L (Q)**
Correct match: (A)–(R); (B)–(S); (C)–(P); (D)–(Q)
Final Answer: 3

64)

Number of atoms = n × N_A × Atomicity
$= 0.1 \times 6.022 \times 10^{23} \times 3$
$= 1.8066 \times 10^{23}$

65)

Mass % = $\frac{\text{Atomic weight} \times 100}{\text{Minimum molecular weight}}$
$4 = \frac{32 \times 100}{M_{min}}$
M_{min} = 800

66) $M = \frac{n_{solute}}{V(L)}$
$0.1 = \frac{0.1}{V}$
V = 1 L = 1000 mL

67) 2 mole of Ca produces 2 mole of CaO
$= 2 \times (40 + 16) = 2 \times 56 = 112$ g

68) **Question Explanation :**
We are asked to identify the hydrocarbon that produces 400 mL of CO_2 and 500 mL of H_2O vapor when 100 mL of it is combusted.

Concept :
Combustion Stoichiometry & Gay-Lussac's Law of Combining Volumes

Solution :
General combustion: $C_xH_y + \left(x + \frac{y}{4}\right) O_2 \rightarrow xCO_2 + \frac{y}{2}H_2O$ Given: 100 mL hydrocarbon → 400 mL CO_2 and 500 mL H_2O.
Ratio: $\frac{400}{100} = 4 \Rightarrow x = 4\frac{500}{100} = 5 \Rightarrow \frac{y}{2} = 5 \Rightarrow y = 10$ So the hydrocarbon is C_4H_{10}.

Final Answer : Option (4)

69) $n_{H_2} = \frac{64}{2} = 32\,mol$

$n_{O_2} = \frac{64}{32} = 2\,mol$

1 mol of O_2 requires = 2 mol of H_2
2 mol of O_2 requires = 4 mol of H_2
Since H_2 is present in excess, therefore, O_2 is the limiting reagent
(1) Wrong
(2) So, O_2 is the limiting reagent
(3) 2mol of O_2 = 4 moles of H_2O = 4 × 18
= 72 g H_2O
Moles of H_2 left = 32 – 4 = 28 mol
= 28 × 2 = 56 g H_2
Mixture contains = (72 g H_2O + 56 g H_2)

70)

Number of atoms = n × N_A × Atomicity
= 1 × 6.02 × 10^{23} × 4
= 24.08 × 10^{23}

71) Volume of C_3H_8 = xL & volume of C_4H_{10}
= (3 – x) L
$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$
1L ———— 3L
xL ———— 3xL
$C_4H_{10} + \frac{13}{2}O_2 \rightarrow 4CO_2 + 5H_2O$
1L ———— 4L
(3 – x)L ———— 4 × (3 – x)L
Volume of CO_2 = 3x + 4(3 – x) = 10

$3x - 4x + 12 = 10$
$x = 2$

72) Read notes

73)

Explain the question : We are asked to calculate the percentage purity of a $CaCO_3$ sample given that 150 g of sample produces 44 g of CO_2 on heating
Concept : This question is based on
Decomposition reaction
$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$
Molar masses $CaCO_3 = 100\ gmol^{-1}$. $CO_2 = 44g\ mol^{-1}$
percentage purity can be calculated using the formula
% Purity = (Actual mass of CO_2 evolved / Theoretical mass of CO_2 from sample) × 100
Solution :
Theoretical mass of CO_2 from 150 g pure $CaCO_3$
(44 ÷ 100) × 150 = 66 g
Actual CO_2 evolved = 44g
Percentage purity = (44 ÷ 66) × 100 = 66.67%
Final answer : option (3)

74)

C	:	H	:	O
$\frac{40}{12}$	:	$\frac{6.7}{1}$	:	$\frac{53.3}{16}$
3.33	:	6.7	:	3.33
1	:	2	:	1

Impirical formula → CH_2O

75) **Explain Question :** We need to find relation between molarity(M) and molality(m) with the help of density and M_{solute} = molar mass of solute.
Concept : This question is based on relation between molarity and molality.
Explanation : $m = \frac{1000 \times M}{(1000 \times d) - (M \times M_{solute})}$
Final answer : 4

76)

Question Explanation: Check correctness of Statement-I and Statement-II for equilibrium: $H_2O(s) \rightleftharpoons H_2O(\square)$
Concept: Le-Chatelier's principle
Solution:
• Melting of ice is **endothermic**.
• Ice has **greater volume** than liquid water.
• Increase in temperature favors **endothermic** direction.

• Increase in pressure favors the side with **smaller volume**.
Statement-I: Increasing temperature favors melting and increasing pressure favors liquid water (smaller volume) → **more water forms → correct**.
Statement-II: Forward reaction is endothermic (**true**), but it says volume of water is greater than ice (**false**, ice has greater volume). Hence overall **incorrect**.

Statement I is correct but Statement II is incorrect.
Final Answer: (Option 2)

77) **Question Explanation :-** Relation Kp and Kc for given reactions is to be found and Match the following columns.
Concept :- $K_p = K_c(RT)^{\Delta n_g}$
K_p = Equilibrium Constant in terms of Pressure
K_c = Equilibrium Constant in terms of Concentration.
R :- Universal gas constant.
T :- temperature
Δn_g = (Moles of gaseous Product) - (Moles gaseous reactant)
Solution :
Rxn :- $C(s) + CO_2(g) \rightleftharpoons 2CO(g)\ ;\ \Delta ng = 2 - 1 = 1$
$K_p = K_c(RT)$
Rxn :- $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)\ ;\ \Delta ng = 2 - 2 = 0$
$K_p = K_c(RT)^0$
Kp = Kc
Rxn :- $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)\ ;\ \Delta ng = 2 - 4 = -2$
$K_p = K_c(RT)^{-2}$
Rxn :- $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)\ ;\ \Delta ng = 2 - 3 = -1$
$K_p = K_c(RT)^{-1}$
Final Answer: option (1)

78) On adding Cl_2 equilibrium shift in forward direction. Addition of inert gas does not alter conc./partial pressure of any component.
So there is no shift in equilibrium.

79) **Explain Question:**
To find out unit of K_c for given reaction.

Concept:
Unit of K_c.

Solution:
4A + 5B ⇌ 4x + 6y
Δn = (4 + 6) – (4 + 5) = 1
unit of $K_c = (M)^{\Delta n}$
$= (M)^1 = \frac{mol}{L} = mol\,L^{-1}$
Final answer: Option (4)

80) **Question Explanation -** Find equilibrium constant for combined reversed reaction

Concept - Multiplication and inversion of equilibrium constants.
Solution -
$A \rightleftharpoons B$, $K_1 = 2$
$B \rightleftharpoons C$, $K_2 = 0.01$
Add reactions: $A \rightleftharpoons C$
$K = K_1 \times K_2 = 2 \times 0.01 = 0.02$
Reverse it: $C \rightleftharpoons A \Rightarrow K = 1 / 0.02 = 50$
Multiply by 2: $2C \rightleftharpoons 2A \Rightarrow K = (50)^2 = 2500 = 2.5 \times 10^3$
Final Answer - Option (3)

81)

Explaining the Question :- Identify factor that changes the equilibrium constant K_c.
Concept:- This question is based on
• K_c depends only on temperature.
Solution :- Only increasing temperature changes K_c.
• addition of inert gas, catalyst, or changing concentrations → affect equilibrium position but not K_c.
Final Answer :- 4

82) **Explain the question :** The question asks to identify the reaction from the given options for which the ratio of the equilibrium constants, $\frac{K_p}{K_c}$, is equal to one. **Concept :** This question is based on relationship between K_p and K_c.

$$\Delta n_g = 0, \frac{k_p}{k_c} = 1$$

Solution : The relationship between K_p and K_c is given by the equation:
$K_p = K_c(RT)^{\Delta ng}$
Δn_g = (Total moles of gaseous products) – (Total moles of gaseous reactants)
Let's calculate Δn_g for each reaction:
1. $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g) \Delta n_g = (1 + 1) – 1 = 1$
2. $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \Delta n_g = 2 – (1 + 3) = –2$
3. $SO_2(g) + 1/2\ O_2(g) \rightleftharpoons SO_3(g) \Delta n_g = 1 – (1 + 1/2) = –0.5$
4. $H_2(g) + I_2(g) \rightleftharpoons 2HI(g) \Delta n_g = 2 – (1 + 1) = 0$
Only in reaction 4 is the value of Δn_g equal to 0.
Final answer : 4

83) **Explain the question :** For which reaction is the relation

$$\log \frac{K_c}{K_p} + \log RT = 0$$

true?
Concept : This question is based on $K_p = K_c(RT)^{\Delta n_g}$
Solution :
• $\log \frac{K_c}{K_p} = -\Delta n_g \log RT$
• Given $\log \frac{K_c}{K_p} + \log RT = 0 \Rightarrow -\Delta n_g \log RT + \log RT = 0 \Rightarrow \Delta n_g = 1$

Check each reaction (Δn_g = moles of gaseous products – reactants):
1. $2A(g) \rightleftharpoons B(g) + 2C(g) \rightarrow \Delta n_g = 3 - 2 = 1$
2. $2A(g) \rightleftharpoons B(s) + 2C(g) \rightarrow \Delta n_g = 2 - 2 = 0$
3. $2A(s) \rightleftharpoons B(g) + C(g) \rightarrow \Delta n_g = 1 - 1 = 0$
4. $A(g) \rightleftharpoons B(g) + C(g) \rightarrow \Delta n_g = 1 - 2 = -1$

Final answer : 1

84)

Explain Question : The question asks for the temperature and pressure conditions that favour the formation of graphite from diamond in the given reaction.
Concept : This question is based on Le chatelier's principle
Solution :

$C(diamond) \rightleftharpoons C(graphite)$; ΔH = –Ve (density ↑) (density ↓)
(volume ↓) (volume ↑)
-Reaction is exothermic
T↓ ⇒ Equilibrium will be shifted in forward direction.
-To favour the side with a large volume, the pressure must be lawered.
-low temperature and low pressure ⇒ C graphite ↑
Final answer : 4

85)

Explain the Question : We are asked to calculate the active mass of 50 g of $CaCO_3$ in a 1L vessel.
Concept : This question is based on active mass.
Solution : The active mass of a pure solid (like $CaCO_3$) is taken as unity (1), regardless of the amount present.
Final answer : 3

86) **Question Explanation :** This question asks which direction a reaction favors when its equilibrium constant K_C, is much greater than 1.
Concept : The K_C expresses the ratio of products to reactants at equilibrium.
Solution : $K_C >> 1$. signifies that the concentration of products is significantly higher than that of the reactants at equilibrium. This means the reaction has progressed extensively to the right, favoring the formation of products. Therefore, the reaction favors the forwards direction.
Final Answer : Option (2)

87) **Question Explanation:** We are to match Column-I with Column-II

	Column-I		**Column-II**
a	Q=K	p	Reaction is nearer to completion
b	Q<K	q	Reaction is not at equilibrium

c	Q>K	r	Reaction proceeds in forward direction
d	K>>1	s	Reaction is in equilibrium
		t	Reaction proceeds in backward direction

Concept: This question is based on reaction quotient (Q) and equilibrium constant (K).
Solution:
• When Q = K, reaction is in equilibrium → (a - s)
• When Q < K, reaction is not at equilibrium and proceeds in forward direction → (b - q, r)
• When Q > K, reaction is not at equilibrium proceeds in backward direction → (c - q, t)
• When K >> 1, reaction is nearer to completion → (d - p)

Final Answer: Option (1)

88) **Explanation:**

Reaction: $NH_3(g) \rightleftharpoons \frac{1}{2} N_2(g) + \frac{3}{2} H_2(g)$ $K_C = 4$
Reverse reaction: $N_2(g) + 3H_2(g) \rightleftharpoons 2\ NH_3(g)$ K_C'
Concept: Relation between K_p and K_p
Solution:
The relation between Kp and Kc is given by:

$K_C' = \frac{1}{(K_C)^2} = \frac{1}{(4)^2}$

$K_P = K_C(RT)^{\Delta n_g}$

$= \frac{1}{(4)^2}(RT)^{-2}$

$K_P = \left(\frac{1}{4 \times 800\ R}\right)^2$

Answer is option (3).

89) $\frac{K_p}{K_c} = (RT)^{\Delta n_g} = 3$; $\Delta n_g = 1$

$0.0821 \times T = 3$

$T = \frac{3}{0.0821} = 36.54$ K

90) $2NaHCO_3(s) \rightleftharpoons Na_2CO_3(s) + CO_2(g) + H_2O\ (g)$

t = 0 0 atm

t = eq 0 0 p_{CO_2} p_{H_2O}

$p_{CO_2} = p_{H_2O} = p$

$2p = 1.04$ $K_P = p_{CO_2} \cdot p_{H_2O}$

$p = 0.52$ atm $K_P = 0.52 \times 0.52$

$= 0.2704$

BIOLOGY

91) **Mast Cells:** These cells are part of the immune system and are involved in allergic reactions and inflammation. They are typically found in connective tissues, such as the dermis of the skin and the lining of the respiratory and digestive tracts.
Answer - 1

92)

Question Explanation: Identify which among the options is a type of loose connective tissue.

Concept: Connective tissue

Solution: The correct answer is option: 1.

A. Areolar tissue is a classic example of loose connective tissue, providing support, elasticity, and cushioning.

Final answer: Areolar

93)

Question Explanation: Identify the type of cartilage in trachea, larynx, and bronchi.

Concept: Types of cartilage

Solution: The correct answer is option: 3

A. Hyaline cartilage forms the tracheal rings, laryngeal cartilages, and bronchial supports, providing rigidity and slight flexibility.

Final answer: Hyaline

94) **Fibroblasts:** These cells are the main producers of collagen and elastin fibers in the extracellular matrix of connective tissues. They play a crucial role in wound healing and tissue repair.
Answer - 2

95)

Question Explanation: Identify which cell type is not found in areolar connective tissue.

Concept: Connective tissue

Solution: The correct answer is option: 3.

A. Schwann cells are part of the nervous system, not connective tissue.

B. Areolar tissue contains macrophages, plasma cells, fibroblasts, mast cells, adipocytes, etc.

Final answer: Schwann cell

96)

Question Explanation: Identify the cartilage type needed to repair the nasal septum.

Concept: cartilage

Solution: The correct answer is option: 3

A. The nasal septum is primarily made of hyaline cartilage, which provides rigidity and support.

B. Recovery or repair of the septum requires hyaline cartilage.

Final answer: Hyaline cartilage

97) **Ligaments:** These are strong bands of connective tissue that connect bones to bones, providing stability and limiting the range of motion at joints.
Answer - 3

98) NCERT Pg. # 101

99) NCERT Pg. # 104

100) NCERT Pg. # 101

101) NCERT Pg. # 102

102) NCERT Pg. # 101, 102

103)

NCERT-XI, Pg. # 103

104)

NCERT-XI, Pg. # 104

105)

NCERT-XI, Pg. # 103

106)

NCERT-XI, Pg. # 102

107)

NCERT-XI, Pg. # 104

108)

NCERT-XI, Pg. # 104

109)

NCERT-XI, Pg. # 102

110)

NCERT-XI, Pg. # 103

111)

NCERT-XI, Pg. # 102, 103, 104

112)

NCERT-XI, Pg. # 104

113)

NCERT-XI, Pg. # 104

114) NCERT XI Pg.# 103, last para

115)

NCERT-XI, Pg. # 103

116)

NCERT, Pg. # 42, 43, 44, 45

117)

NCERT, Pg. # 41 to 46

118) NCERT, Pg. # 46, 47

119) NCERT, Pg. # 45

120) NCERT, Pg. # 40

121)

NCERT, Pg. # 41

122)

NCERT, Pg. # 44

123)

NCERT, Pg. # 39

124)

NCERT-XI Pg. # 42, 43, 44

125)

NCERT, Pg. # 44

126)

NCERT, Pg. # 43

127) NCERT XI, Pg # 56

128)

NCERT, Pg. # 39

129)

NCERT, Pg. # 44

130)

NCERT, Pg. # 51

131)

NCERT, Pg. # 43

132)

NCERT XI, Pg.no. 56

133)

NCERT, Pg. # 46

134) NCERT Pg. # 49, 52, 58

135)

NCERT, Pg. # 46

136)

Pg. No. 53, 54_XI NCERT

137) **Explaining:** Identify the freshwater bony fishes from the given options
Concept: This question is based on Distinguish between freshwater and marine fish, and bony vs. cartilaginous fish.
Solution:
Rohu and Clarias (Magur) are common freshwater bony fishes.
Other options include marine or non-bony fishes (e.g., flying fish, sea horse, saw fish).

138)

Explain Question : Asking how many features are not common between Chondrichthyes (shark) and Osteichthyes (sea horse).

Concept : This question is based on Chondrichthyes, Osteichthyes.

Solution : The correct answer is option 2. (three)

(B) Air bladder → Absent in Chondrichthyes, present in Osteichthyes → Not common
(D) Terminal mouth → Chondrichthyes = ventral mouth, Osteichthyes = terminal mouth → Not common
(E) Oviparous → Chondrichthyes = many viviparous, Osteichthyes = mostly oviparous → Not common

(A) Marine → Both groups include marine fishes → Common
(C) Cold blooded → Both are poikilothermous → Common
(F) Presence of scales →Both have scales(don't consider the types of scales) → common

Not common features = B, D, E = Three

Final Answer : option (2)

139)

NCERT, Pg. # 48

140)

Annelida, Arthropoda and Chordates are only metamerically segmented animals.

141)

NCERT-XI, Pg. # 88, 89

142)

NCERT-XI, Pg. # 99

143)

NCERT-XI, Pg. # 98

144)

NCERT-XI, Pg. # 98

145)

NCERT-XI, Pg. # 101

146)

NCERT-XI, Pg.# 95

Explain Question:
The question asks about the main function of the Golgi body among the listed options.
Concept : This question is based on Golgi body.
Explanation :
• The Golgi apparatus is part of the endomembrane system.
• Its primary function is modification, packaging, and secretion of materials (proteins, lipids, polysaccharides).
Final answer (3) : Secretion

147)

NCERT-XI, Pg. # 95

148)

NCERT-XI, Pg. # 96, 97

149)

NCERT-XI, Pg. # 98

150)

NCERT-XI, Pg. # 97

151)

NCERT-XI, Pg. # 100

152)

NCERT-XI, Pg. # 96

153)

NCERT-XI, Pg. # 97

154)

NCERT-XI, Pg. # 96, 97

155)

NCERT-XI, Pg. # 99

156)

NCERT-XI, Pg. # 100

157)

NCERT-XI, Pg. # 98

158)

NCERT-XI, Pg. # 102

159)

NCERT-XI, Pg. # 99

160)

NCERT-XI, Pg. # 101

161)

NCERT-XI, Pg # 89

162)

NCERT-XI, Pg. # 100, 101, 102

163)

SOLUTION —
Interphase is the phase of the cell cycle where the cell prepares for division. It is subdivided into three stages:

A. G_1 (Gap 1) – Cell growth and normal functions.

B. S (Synthesis) – DNA replication occurs.

C. G_2 (Gap 2) – Further growth and preparation for mitosis.

The G_o (G-zero) phase is not technically part of interphase, but it's a resting phase that some cells enter from G1, where they are not actively dividing.

The correct answer is: OPTION(2) G_1, S, G_2.

164) NCERT-XI, Pg. # 96

165) NCERT-XI, Pg. # 101

166)

NCERT-XI, Pg. # 94

167)

NCERT-XI, Pg. # 95

168)

Question Explanation: Identifying the prolonged meiotic stage in oocytes.

Concept: Meiosis in Oogenesis.

Solution:
During diplotene (which is part of prophase I of meiosis), the oocyte undergoes a prolonged resting phase. This phase can last a significant amount of time in many vertebrates, particularly in species where the eggs are held in a state of suspended development until triggered by hormonal signals or environmental factors to resume development.

Final Answer: 3

169) NCERT-XI Pg. # 85, 103

170) NCERT-XI, Pg # 127

171)

NCERT-XI, Pg # 126

172) NCERT-XI, Pg # 120, 121

173) NCERT-XI, Pg # 121

174) NCERT-XI, Pg # 125

175)

NCERT-XI, Pg # 95, 96, 97, 99

176)

NCERT-XI, Pg # 121

177)

NCERT-XI, Pg # 121, 122, 123, 124

178)

NCERT-XI, Pg # 126, 127, 128

179)

NCERT-XI, Pg # 121, 126

180) NCERT-XI, Pg # 126, 127, 128